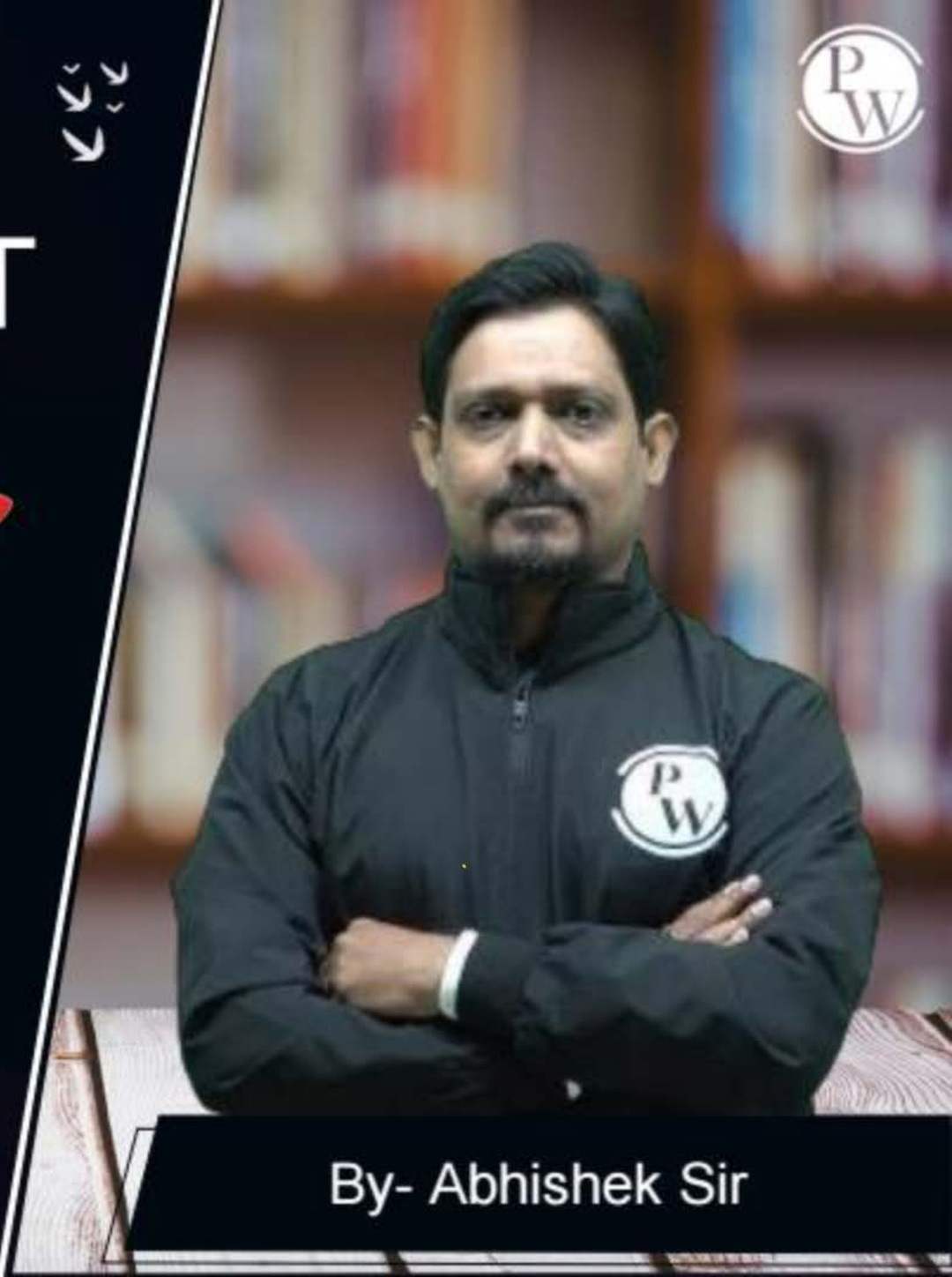
Computer Science & IT



C Programming

String

Lecture No. 01



By- Abhishek Sir

Recap of Previous Lecture



- Topic: Two Dimensional Array
- Topic: String
- Topic
- Topic
- Topic

Slide

Topics to be Covered



- Topic — String
- Topic — String function
- Topic
- Topic
- Topic

2-Dimensional

Slide



GATE 2020

Consider the following C program.

```
# include <stdio.h>
int main (){
int a[4][5] =  {{1, 2, 3, 4, 5},
                {6, 7, 8, 9, 10},
                {11, 12, 13, 14, 15},
                {16, 17, 18, 19, 20}};

printf("%d\n", *(*(a+**a+ 1)+4));
return (0);
}
```

The output of the program is _____.

$a[i][j] = *(*(a+i)+j)$

Homework $a[4][5]$

$a \rightarrow$ Address 1-D array

$*a$ - Address integer

$**a$ integer

$*(*(a+**a+1)+4)$

$*(*(a+1+1)+4)$

$*(*(a+2)+4)$

$a[2][4]$

15

Slide



String

$a[i][j] = *(*(a+i)+j)$

$a[i][j] \rightarrow **$

a : Address
1 D array

*a : Address
integer

**a integer

String

char a[ ] = "string"



termination

printf("%s", a+2); → print - ring

a+2 → 102

printf("%s", a+6); ⇒ print Nothing. (No output)

100+6 = 100+6×1 = 106

print("%s", a)

a → address of s

Slide

String

If string by array character

```
char a[] = "String";
a[1] = 'p';
printf("%s", a);
```

100	101					
S	~~t~~ p	r	i	n	g	\0

Spring output

```
a = a + 1
a
```

Slide

String

```
char a[] = "string";
```

```
a = a + 1;
```

s	t	r	i	n	g	\0	
100	101	102	103	104	105	106	

a | 100 |

Compiler will Not allow update value a

will compiler allow to modify value of a

Slide



String

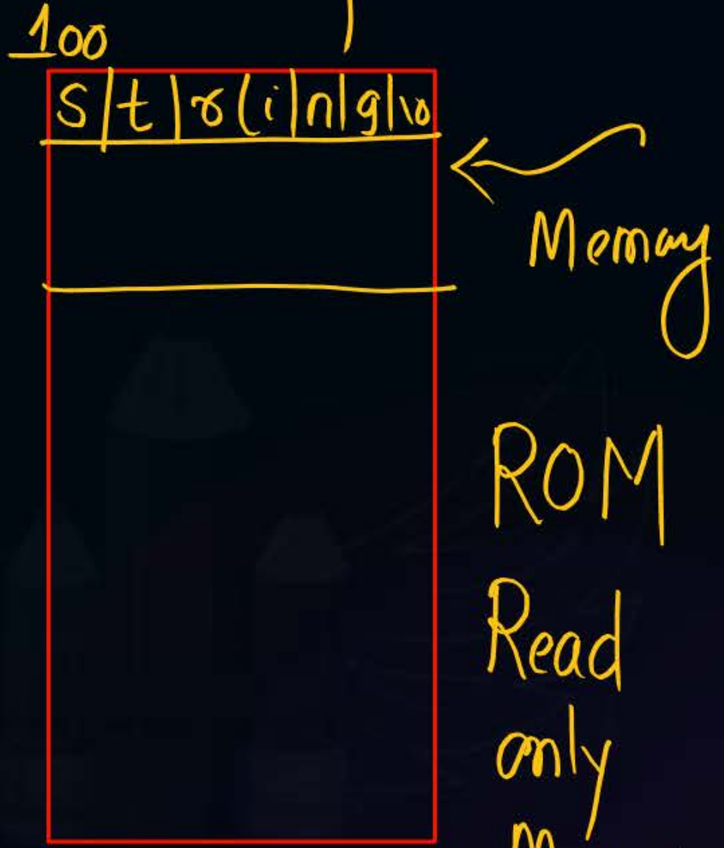
The association of Name of array with the location is Not modifiable

String

Second way of declaring string using character pointer

```
char *ptr = "string";
```

ptr | 100 |



ROM
Read only Memory

ptr = ptr + 1; Is it allowed

yes

ptr[1] = 'p' ⟶ printf("%s", ptr)

Slide



String

if string is maintained by character pointer that string stored in ROM (Read only Memory)

we cant modify

Slide



String

char a[] = "string"

1. a = a + 1 Not allowed

2. a[2] = 'p' allowed

3.

char * ptr = "string";

1. ptr = ptr + 1 allowed

2. ptr[2] = *(ptr + 2)

= 'p'; Not allowed

ROM

Slide

String

3. `char ch1[] = "string";`

`char ch2[] = "string";`

100

s	t	r	i	n	g	\0

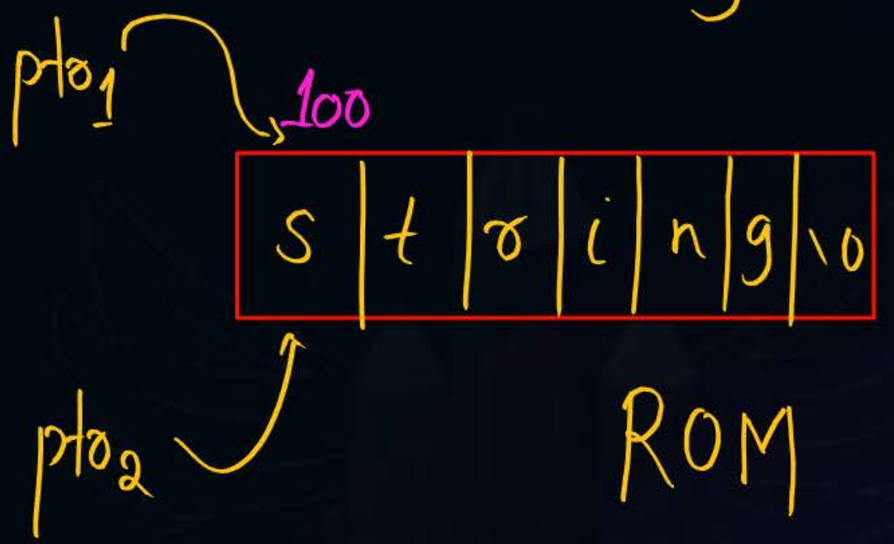
200

s	t	r	i	n	g	\0

3. `char *ptr1 = "string";`

`char *ptr = "string";`

} if Same string.



Slide



String

*(ch+1)

Array of string

char ch[2][10] = { parakram , "vijay" };

ch[0] ch[1]

Question : How to print parakram

ch → Address of 1-D array

*ch → Address of 1 character of 1D array → % address of 1st char

**ch → character

Slide

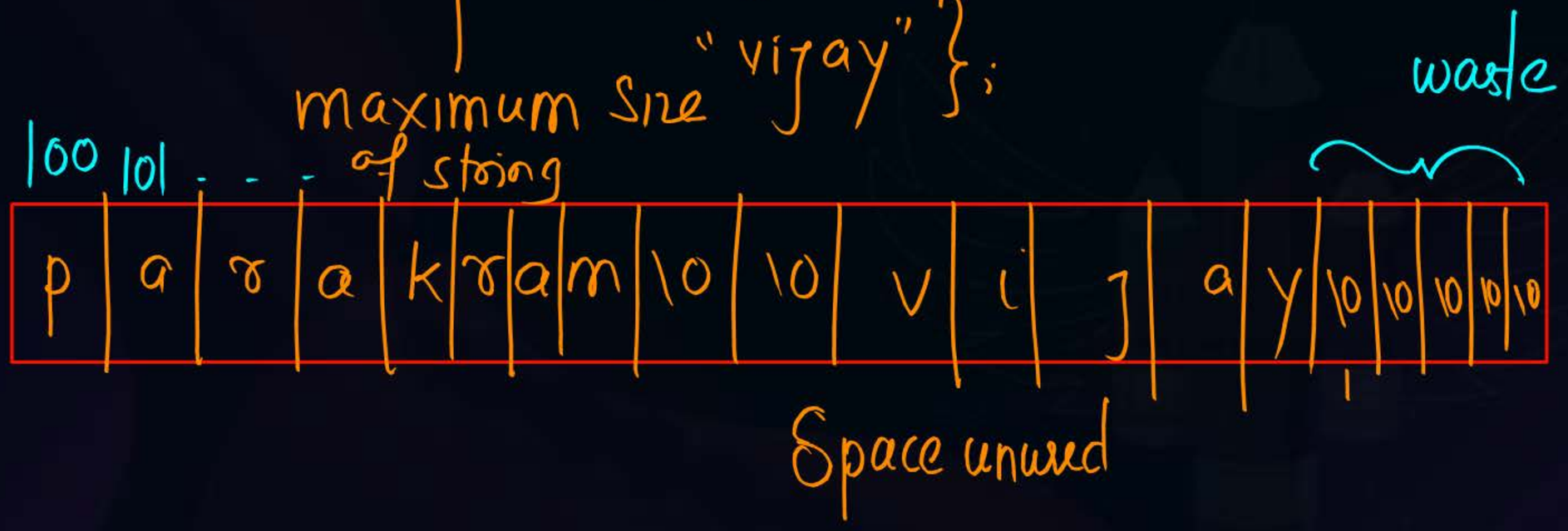


String

printf("%s", ____) Address of first character of string

char ch[2][10] = {"parakram", "vijay"};

maximum size of string



Slide



String

String character pointer

```
char *ptr = "parakram";
char *ptr1 = "vijay";
```

Array of character pointer

```
char *ptr[2] = {"parakram", "vijay"}
```

Read only memory

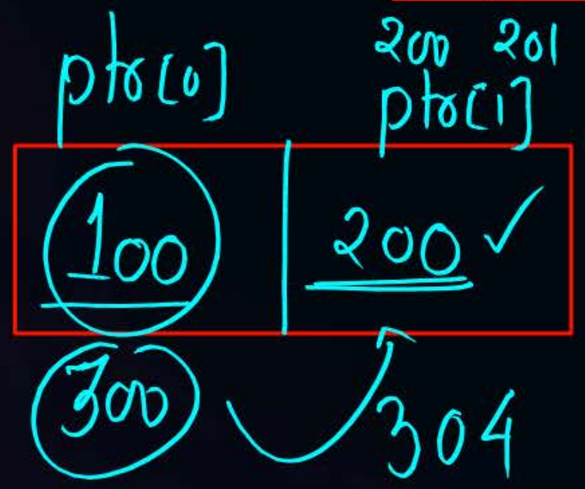
Slide

String

Memory Diagram



String size may be different & Space will Not wastage ptr



array of pointer.

*(ptr+1)

Slide



String

* How to print parakram

ptr[0] = *(ptr+0) = *ptr

printf("%s", *ptr+1);

printf("%s", *(ptr+1));

Slide



String

*ptr + 1

↑ Unary operator

100 + 1 = 101

Slide



String

String.h

strlen( );

it returns length of string

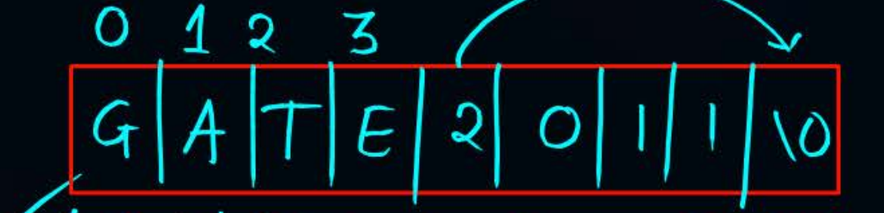
'\0' character is Not counted.

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GATE 2011

Q What does the following fragment of C-program print?

```
char c[]="GATE2011";
char *p =c;
printf("%s", p+p[3]-p[1]);
```



A

2011

(A) GATE2011

(B) E2011

(C) 2011

(D) 011

p 100

100 + 69 - 65

100 + 4 = 104

65-A
66-B
67 C
68-D 69-E

p[3]

*(p+3) = *(103) = E

*(p+1) = *(101)

Slide



2 mins Summary

- Topic — string
- Topic — array of string
- Topic — array of pointer
- Topic
- Topic

Slide

THANK - YOU

